PROJECT COLLECTION

REZA BABAPIRI

# Automotive & Vehicle Simulation

Vehicle dynamics, driver assistance and co-simulation

5 selected documented projects

Project context / Methods / Evidence / Limitations

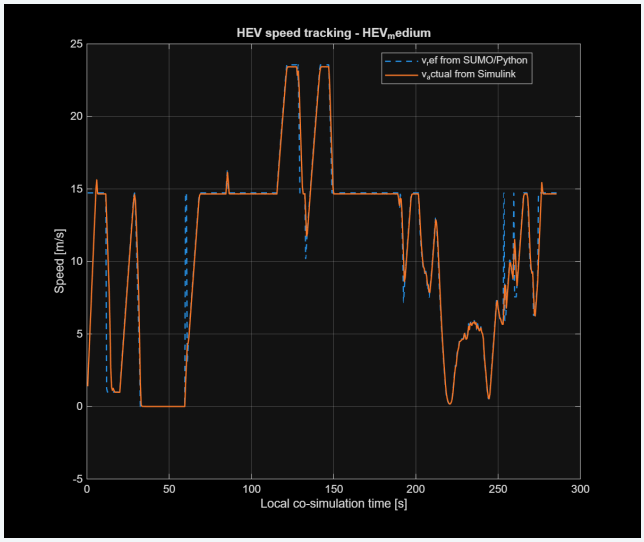
reza-babapiri.github.io

Academic work and anonymised professional experience | Source-derived summaries

AUTOMOTIVE SIMULATION · PYTHON

# EV / HEV Traffic Co-Simulation | Rome

SUMO traffic + Python bridge + MATLAB/Simulink vehicle models




## ENGINEERING SUMMARY

Built a Rome urban-traffic simulation workflow using SUMO route data and Python interfaces linked with MATLAB/Simulink EV and HEV models. The project contains multiple traffic intensities, vehicle-model files and saved scenario results.

## METHOD / CONTRIBUTIONS

- Prepared light, medium and heavy traffic scenarios with SUMO-compatible routes.
- Connected traffic speed traces and vehicle simulation via Python/MATLAB bridging scripts.
- Compared speed tracking, state-of-charge trends and energy/fuel-equivalent outputs across scenarios.

## RESULTS & EVIDENCE

- Archived CSV summaries contain light, medium and heavy scenarios and actual-versus-reference speed records.
- Separate EV and HEV Simulink models and scenario plots are included.

### SCOPE / LIMITATION

Saved scenario results have not been independently rerun in the current environment; do not interpret equivalent-fuel quantities as laboratory vehicle measurements.

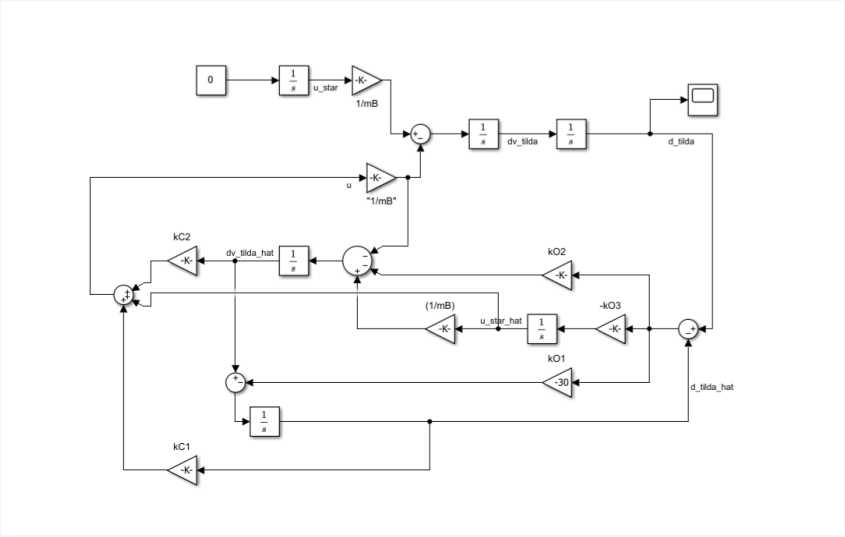
### SUPPORTING RECORD

Project RAR contains Python scripts, MATLAB bridges, .slx models, route files, result CSVs and figures.

ADVANCED DRIVER ASSISTANCE · CONTROL

# Adaptive Cruise Control | PID vs Observer

Distance-error regulation and disturbance estimation




## ENGINEERING SUMMARY

Studied adaptive cruise control as a following-distance regulation problem. Implemented and compared a PID approach against an observer-based feedback strategy to estimate unmeasured dynamics and disturbances.

## METHOD / CONTRIBUTIONS

- Formulated distance-error dynamics with leader acceleration and resistive disturbance terms.
- Designed PID control and a state observer with performance targets from the coursework.
- Compared time-domain responses and sensitivity measures.

## RESULTS & EVIDENCE

- The supplied report compares nominal simplicity of PID control with advantages of observer-based estimation.

### SCOPE / LIMITATION

Controller comparison is simulation-based; it is not a road-validated ADAS system.

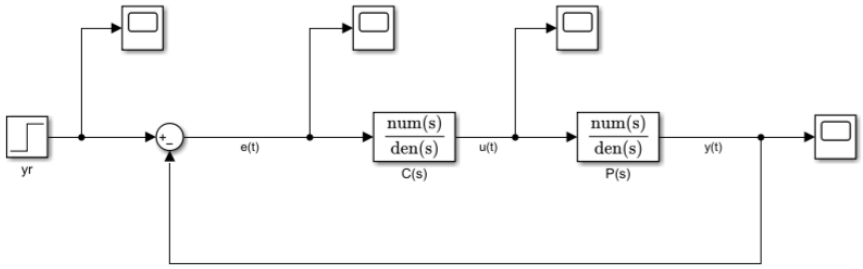
### SUPPORTING RECORD

24-page CMS HW4 part 2 report, MATLAB code and comparison plots.

VEHICLE DYNAMICS · STATE SPACE

# BMW 730 | Steering Dynamics

Speed-dependent stability, damping and steering response




## ENGINEERING SUMMARY

Analysed a BMW 730 bicycle-model representation to understand speed-dependent lateral dynamics, controllability, observability, damping and steering response.

## METHOD / CONTRIBUTIONS

- Evaluated eigenvalues, transfer functions and speed-dependent dynamic properties.
- Compared understeering / oversteering conditions using zero-frequency gains.
- Designed feedback for selected lateral or yaw-rate response objectives.

## RESULTS & EVIDENCE

- The report documents a damping-ratio improvement in a selected closed-loop case (approximately 0.088 to 0.628).

**SCOPE / LIMITATION**

Based on an academic vehicle model, not measurement of a production BMW vehicle.

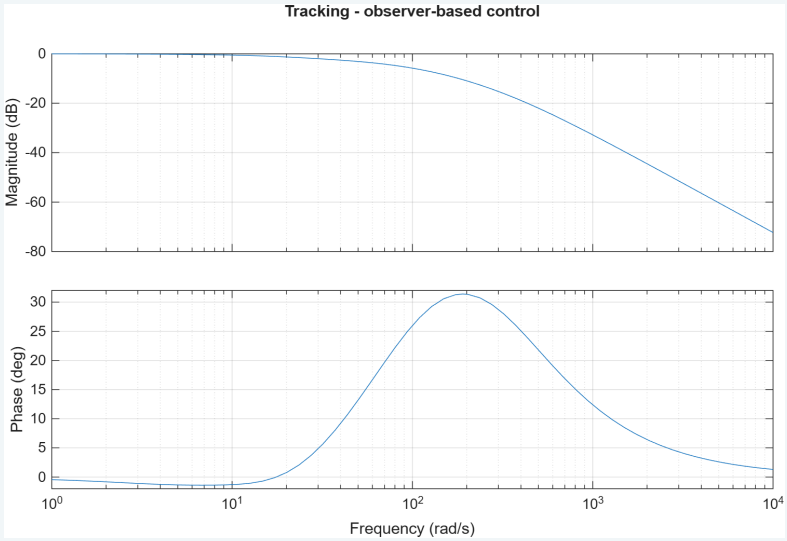
**SUPPORTING RECORD**

38-page CMS HW5 part 1 report and supporting MATLAB code.

STATE ESTIMATION · VEHICLE CONTROL

# Active Steering | Observer & Wiener Filter

PI feedback versus observer-based yaw-rate regulation




## ENGINEERING SUMMARY

Extended the BMW 730 steering model into controller and state-estimator comparisons. Investigated PI pole placement, Luenberger observer design and a Wiener-filter-inspired observer under uncertainty and noise.

## METHOD / CONTRIBUTIONS

- Designed a PI controller using characteristic-polynomial constraints.
- Compared deterministic state observation with covariance-weighted Wiener estimation.
- Examined speed-dependent tracking and stability through time- and frequency-domain results.

## RESULTS & EVIDENCE

- The report analyses state-estimation quality and trade-offs between controller complexity and measurement noise sensitivity.

### SCOPE / LIMITATION

Filter performance depends on assumed process and measurement covariance.

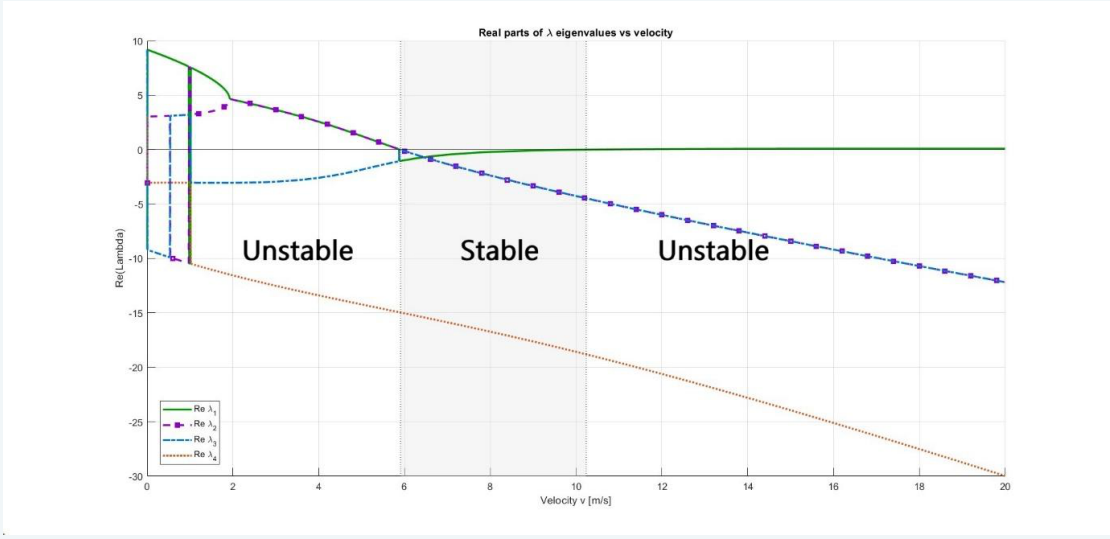
### SUPPORTING RECORD

38-page CMS HW5 part 2 report and MATLAB analysis script.

DYNAMIC SYSTEMS · BALANCE CONTROL

# Bicycle Roll | Stability & Feedback

Speed-dependent roll dynamics and feedback stabilization




## ENGINEERING SUMMARY

Examined simplified roll dynamics and a Whipple-model bicycle representation, exploring why stability and steering authority change with forward speed.

## METHOD / CONTRIBUTIONS

- Derived steering-to-roll transfer functions and investigated poles, zeros and gain.
- Compared proportional stabilization and more capable feedback designs.
- Evaluated speed-dependent stability and limitations near low forward speed.

## RESULTS & EVIDENCE

- The report highlights sensitivity of stabilization to model order, speed and sensing assumptions.

### SCOPE / LIMITATION

Academic model study; no physical autonomous bicycle test is documented.

### SUPPORTING RECORD

26-page CMS HW7 report with MATLAB analysis.